Mar 25, 2026 Version 1

Human Brain Nuclei Isolation Using Sucrose Gradient Ultracentrifugation V.1

DOI
https://dx.doi.org/10.17504/protocols.io.x54v9yrm4g3e/v1

Zhang Lab[1]

[1]Yale University

ASAP Collaborative Res...

N Nick C Buitrago-Pocasangre
Yale University

OPEN ACCESS

DOI: https://dx.doi.org/10.17504/protocols.io.x54v9yrm4g3e/v1

Protocol Citation: Zhang Lab 2026. Human Brain Nuclei Isolation Using Sucrose Gradient Ultracentrifugation. **protocols.io** https://dx.doi.org/10.17504/protocols.io.x54v9yrm4g3e/v1

License: This is an open access protocol distributed under the terms of the **Creative Commons Attribution License**, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited

Protocol status: Working
We use this protocol and it's working

Created: March 27, 2022

Last Modified: March 25, 2026

Protocol Integer ID: 59970

Keywords: Human brain Nuclei, Sucrose gradient, Isolation, Ultracentrifugation, ASAPCRN, isolation of human brain nuclei, human brain nuclei isolation, using sucrose gradient ultracentrifugation, sucrose gradient ultracentrifugation this protocol, human brain nuclei, gradient ultracentrifugation, isolation

Funders Acknowledgements:
ASAP
Grant ID: ASAP-000529

Abstract

This protocol describes about isolation of human brain nuclei by sucrose gradient ultracentrifugation.

Attachments

392-854.docx
20KB

Materials

Reagents

MilliQ water DNase-, RNase-free water

	A	B
	Sucrose Homogenization Buffer	
	Sucrose	2 M
	Hepes (pH 7.9)	10 mM
	KCl	25 mM
	EDTA (pH 8.0)	1 mM
	Glycerol	10%
	Ribonuclease (RNase) inhibitors freshly added	80 U/ml

	A	B
	Nuclei Suspension Buffer	
	Hepes (pH 7.4)	15 mM
	NaCl	15 mM
	KCl	60 mM
	MgCl2	2 mM
	CaCl2	3 mM
	RNase inhibitors freshly added	80 U/ml)

Materials

- Eppendorf tube
- surgical blade
- Ultracentrifuge
- SW28 rotor
- haemocytometer slide (C-Chip, DHCN015)

38.5 mL Open-Top Thinwall Ultra-Clear Tube 25 × 89mm - 50Pk **Beckman Coulter Catalog #**344058

Troubleshooting

Isolation of Human Brain Nuclei

1h 10m

1 Take brain tissue sample out of -80 °C freezer and put on dry ice. Transfer the sample from the bag or tube to a petri dish and carefully cut a small piece off with a surgical blade, keeping it on dry ice. Replace the remaining piece in the bag and put the small piece in a labeled Eppendorf tube. Keep the Eppendorf tube on dry ice. Repeat for all samples, using a different petri dish and surgical blade for a new sample.

2 Optional: Weigh Eppendorf tubes if needed and record weights. 50 mg per sample.

3 Pour 25 mL per sample of sucrose homogenization buffer into 50 mL Falcon tube and place On ice. Add 1 µL of RNase Inhibitor (40 U/µL) per mL of homogenization buffer. Shake tube to mix.

4 Pipette 15 mL homogenization buffer with RNase Inhibitor into 15 mL Dounce tissue grinder On ice. Add tissue piece and homogenize.

Note

Note: Slowly break big tissues (first 3-4 strokes up and down), then dounce up and down 10 times with the “loose” pestle followed by 10 times with the “tight” pestle.

Note: Pull the pestle out slowly to avoid spilling. Each sample takes 5-10 minutes.

5 Label bottom of 50-mL ultracentrifuge tube (in case it breaks in the centrifuge). Pipette 10 mL fresh homogenization buffer to the tube as a cushion, and carefully pour the 15 mL of homogenate on top of it to maintain two separate layers.

6 Place the ultracentrifuge tubes in the swing buckets. Weigh and balance the buckets containing the samples by adding a few drops of fresh homogenization buffer to the top.

7 Ultracentrifuge at 25000 rpm, 4°C, 01:00:00 in SW28 rotor for ultracentrifuge. 1h

Note

- Do not use a fixed angle rotor. Always use a swinging bucket.
- Thin wall 50-mL tube: BECKMAN 344058 (Tube, Thinwall, Ultra-Clear™)
- **Wash glassware**: Rinse each piece with tap water 10 times. Then rinse with MilliQ water 5 times. Lay out to dry on paper towels. Once dry, rinse one time with DNase-, RNase-free water, air dry and store for next time.

8 Add 2 µL RNase Inhibitor (40 U/µL) per mL of nuclei suspension buffer, making 5 mL total buffer per sample. Mix and place On ice .

9 Remove all the supernatant by pouring off into a petri dish. Place upside-down in petri dish for a few seconds. Cut off the wall of the thin centrifuge tube with a clean razor blade.

10 Resuspend the nuclei pellet at the bottom of the cut ultracentrifuge tube in 500 µL nuclei suspension buffer. Pipette gently up and down in the solution 10–20 times. Transfer this buffer into a labeled Eppendorf tube and repeat with an additional 500 µL nuclei suspension buffer. Place On ice .

11 In a new labeled Eppendorf tube, make a 1:1 dilution of the nuclei suspension and Trypan Blue. Count nuclei (round blue dots) using haemocytometer slide (C-Chip, DHCN015). Count each 4×4 grid at the corners and average the counts.

12 Concentration of the nuclei: Average count x 2 (dilution in Trypan Blue) x 10^4/mL.

13 Spin down nuclei at 800 x g, 4°C and 9 acceleration/deceleration for 00:10:00 . 10m

14 Carefully remove supernatant with pipette.

Note

Do not disturb the pellet at the bottom.

Resuspend in nuclei suspension buffer: the ideal concentration for 10X is 1000 nuclei/µL.

15 Make a new 1:1 dilution of the nuclei suspension and Trypan Blue. Recount nuclei to ensure the concentration is adequate for 10x (700-1200 nuclei/µL). Ideally, there would be 50 nuclei per 4×4 grid in the haemocytometer slide.

16 Submit samples for 10x Genomics Chromium loading (single cell 3′ V3.1 chemistry).